AAEE CASE REPORT #10:

Alcoholic polyneuropathy is a common disorder, characterized by nonspecific clinical, electromyographic, and pathologic features. Nerve conduction studies and needle electromyography disclose findings consistent with a generalized, sensorimotor, axonal degenerative, polyneuropathy. Alcoholic polyneuropathy may be suspected when such findings occur in the setting of long-standing alcoholism and malnutrition. The lack of specific diagnostic criteria for alcoholic polyneuropathy requires that other potential etiologies be excluded before that diagnosis is made. The institution of a nutritionally-balanced diet constitutes the principal therapy for alcoholic polyneuropathy, however, significant improvement following such therapy is not frequently seen.

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ALCOHOLIC POLYNEUROPATHY

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INTRODUCTION

Alcoholic polyneuropathy is one of the most common forms of peripheral nerve disease. In one series it accounted for nearly 30% of all cases of generalized polyneuropathy.[14] Although the precise incidence of alcoholic polyneuropathy is unknown, it has been found in approximately 9% of alcoholic patients requiring hospital admission for alcohol-related disorders.[28] When one considers the high prevalence of asymptomatic alcoholic patients with physical signs of polyneuropathy[28,29] or abnormalities of nerve conduction,[3,6,13,17,31,33] the magnitude of the problem of alcoholic polyneuropathy becomes apparent. The following case report illustrates some of the typical clinical and electromyographic features of alcoholic polyneuropathy.

CASE REPORT

History. The patient was a 53-year-old white woman with a long-standing history of serious alcoholism who presented with complaints of a pins and needles sensation in her feet and burning pain and weakness in her legs. The sensory symptoms had begun insidiously 1 year earlier with nocturnal aching in the feet and legs. Over the ensuing months the paresthesias and burning dysesthesias evolved in a slowly but steadily progressive fashion. Three months prior to admission the patient began to notice generalized weakness in her legs such that she experienced progressive difficulty walking and climbing stairs. There were no other pertinent medical or neurologic complaints.

Past medical history was notable for chronic alcoholism since early adult life. The patient admitted to drinking four to five beers and two to three shots of liquor per day for the past several years. Her nutritional history disclosed an inadequate diet and a weight loss of 25 pounds. This weight loss, which occurred over the preceding year, represented a 25% loss from her baseline weight of 95 pounds. The patient had smoked one pack of cigarettes per day for the prior 30 years. There was no history of diabetes mellitus, toxic exposure, or chronic medical illnesses. Family history was negative for peripheral nerve disease and other neurologic disorders.

Physical Examination. General physical examination disclosed blood pressure 140/90mm Hg, pulse 82/min, respirations 16/min, and temperature 36°C. The patient was thin and cachetic. Other pertinent findings included several spider angiomata over the neck and chest, mild palmar erythema, and mild hepatomegaly. Neurologic examination disclosed mild to moderate weakness and atrophy of all muscles below the knees. Muscle power in the upper extremities was normal. Deep tendon reflexes were absent at the Achilles tendons, diminished at the patellar and brachioradialis tendons bilaterally, and normal at the biceps and triceps tendons bilaterally. Plantar re-

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sponses were downgoing bilaterally. Vibratory sensation was absent at the toes and diminished at the ankles, knees and finger tips bilaterally. Perception of both pin and light touch was decreased below the knees but was preserved in the upper extremities. Cerebellar function, gait and station were unremarkable.

Laboratory. A blood count disclosed a hemoglobin of 11.0 gm/dl, a hematocrit of 33.5%, an MCV of 112 μm^3, a WBC of 9,500/mm^3, and a platelet count of 280,000/mm^3. Serum folate was undetectable but serum B12 was normal as were serum electrolytes, BUN, and glucose. Serum glutamic oxaloacetic transaminase was slightly elevated at 44 IU/L (normal 7-40 IU/L) but other liver function tests were normal. The chest x-ray and electrocardiogram disclosed no abnormalities.

Electromyography and Nerve Conduction Studies. *Methods:* Standard techniques were used for the nerve conduction studies and the concentric needle examination.[15] Antidromic sensory conduction studies were performed on the sural, superficial peroneal, median, ulnar, and radial nerves. Motor conduction studies and F waves were obtained on the peroneal, tibial, median and ulnar nerves. H reflexes were obtained bilaterally by stimulating the tibial nerve at the popliteal space and recording over the soleus. Concentric needle electromyography was performed on several muscles in the right upper and lower extremities (Table 2).

Results: The results of the nerve conduction studies are detailed in Table 1. Significant abnormalities noted included absent sural and superficial peroneal sensory responses, a low motor amplitude response from the peroneal nerve, slowed motor conduction velocities and prolonged F wave latencies of the peroneal and tibial nerves, and absent H reflexes. In the upper extremity motor conduction velocities from the median and ulnar nerves were mildly slowed and the F wave latency of the ulnar nerve was mildly prolonged.

Findings on the concentric needle examination are shown in Table 2. Insertional activity was increased in all distal muscles in the lower extremity. Fibrillation potentials and positive sharp waves were recorded in moderate numbers in the extensor digitorum brevis and flexor digitorum longus muscles and in mild numbers in the tibialis anterior and medial gastrocnemius muscles. Moderate numbers of increased-duration polyphasic motor unit potentials (MUP) were recorded in the distal muscles of the lower extremity. In this same distribution a mild reduction in the interference pattern characterized by a reduced number of MUPs firing at rapid rates during attempted maximum contraction was noted. In the distal muscles of the upper extremity a few increased-duration MUPs were noted but no abnormal spontaneous activity was detected.

Interpretation: The constellation of findings on the nerve conduction studies and needle electromyography is characteristic of a generalized, sensorimotor, axonal degenerative polyneuropathy. These findings, though nonspecific for etiology, are quite consistent with alcoholic polyneuropathy.

Table 1. Nerve conduction studies.

Nerve stimulated	Recording site	Amplitude uV (S) or mV (mt)	Distal latency msec	Conduction velocity M/sec
Sural-S	ankle	No response		
Sup Per-S	ankle	No response		
Peroneal-Mt	EDB	2.0 (>2.5)	5.9 (<6.0)	33 (>40)
Tibial-Mt	AH	8.0 (>4.0)	5.8 (<5.8)	31 (>40)
Peroneal-F	EDB		57.0 (<55.0)	
Tibial-F	AH		62.0 (<57.0)	
R H-Reflex	soleus	No response		
L H-Reflex	soleus	No response		
Median-S	index	15.0 (>15.0)	3.5 (<3.6)	
Ulnar-S	5th digit	18.0 (>10.0)	3.1 (<3.1)	
Radial-S	thumb	25.0 (>14.0)	2.4 (<2.7)	
Median-Mt	APB	12.0 (>6.0)	3.5 (<4.0)	48 (>51)
Ulnar-Mt	ADM	7.5 (>7.0)	2.8 (<3.0)	45 (>50)
Median-F	APB		28.9 (<31.0)	
Ulnar-F	ADM		32.2 (<31.0)	

Sup Per = superficial peroneal, S = sensory, Mt = motor, F = F wave, R = right, L = left, EDB = extensor digitorum brevis, AH = abductor hallucis, ADM = abductor digiti minimi, APB = abductor pollicis brevis. Normal values are in parentheses.

Table 2. Concentric needle exam.

Muscles	Insertional activity	Spont act Fibs-Psw	Motor number	Unit recruit	Potentials morph
Extensor digitorum brevis	increased	many	↓ ↓	rapid	many IDAP
Flexor digitorum longus	increased	many	↓ ↓	rapid	many IDAP
Tibialis anticus	increased	few	↓	rapid	few IDAP
Medial gastrocnemius	increased	few	↓	rapid	few IDAP
Vastus lateralis	normal	none	normal	normal	normal
Gluteus medius	normal	none	normal	normal	normal
Low lumbar paraspinals	normal	none	normal	normal	normal
First dorsal interosseous	normal	none	normal	normal	few IDAP
Flexor pollicis longus	normal	none	normal	normal	few IDAP
Pronator teres	normal	none	normal	normal	normal
Biceps	normal	none	normal	normal	normal
Triceps	normal	none	normal	normal	normal

Fibs-Psw = fibrillation potentials and positive sharp waves, recruit = recruitment pattern, morph = morphology, ↓ = mild decrease in number, ↓ ↓ = moderate decrease in number, IDAP = increased duration and increased amplitude polyphasic motor unit potential.

Hospital Course. The patient was treated with multivitamins and her folate deficiency anemia gradually improved. Despite correction of her underlying nutritional deficiency state only mild improvement of her symptoms and signs of polyneuropathy was noted.

DISCUSSION

Although the association between alcoholism and peripheral nerve disease may have been first recognized in the eighteenth century,[27] it was not until the early twentieth century that the relationship between alcoholic polyneuropathy and nutritional deficiency was first proposed.[20,22,32] These writings laid the foundation for the modern concept that alcoholic polyneuropathy is a disorder of nutritional deficiency, indistinguishable from neuropathic beriberi.[27]

Alcoholic polyneuropathy invariably occurs in the setting of longstanding serious alcoholism and the secondary nutritional deficiency caused by the displacement of essential nutrients from the diet by alcohol.[27] The majority of patients with alcoholic polyneuropathy report substantial weight loss (often in the range of 30–40 lbs) prior to or concomitant with the evolution of the peripheral neuropathy.[28,29] Although some patients with alcoholic polyneuropathy, particularly those who drink beer, may not demonstrate weight loss and may even appear obese, evidence of nutritional deficiency is invariably present.[27] A disproportionately high incidence of alcoholic polyneuropathy has been observed in women.[28] There is no adequate explanation for this curious fact.

The clinical features of alcoholic polyneuropathy conform to those of a generalized, symmetric, sensorimotor, distal axonopathy.[27–29] Usually the polyneuropathy develops insidiously; however, a rapidly progressive course is sometimes noted. Most patients present with weakness, paresthesias, or pain in the feet or legs. Often the pain takes the form of a distressing burning sensation or a sharp lancinating pain. Physical signs include distal sensory loss, diminished or absent deep tendon reflexes, and distal atrophy and weakness. In some cases the proximal leg muscles are weaker than the distal muscles, giving the false impression of a chronic alcoholic myopathy.[10,16] Rarely, and only in severe cases, the vagus nerve is clinically involved producing dysphonia and dysphagia.[21] Subclinical involvement of the autonomic nervous system however, is common.[8] Frequently, asymptomatic alcoholic patients are found to have distal sensory loss, absent reflexes or distal weakness and wasting, indicating the presence of an underlying polyneuropathy.[28]

Other common problems resulting from chronic alcoholism and nutritional deficiency are often present in patients with alcoholic polyneuropathy.[14,28,29] These include a variety of skin lesions, macrocytic anemia, fatty hepatosis and more serious disorders such as Wernicke-

Korsakoff disease, cerebellar degeneration, or decompensated alcoholic liver disease.

Early pathological studies of alcoholic polyneuropathy disclosed both segmental demyelination and axonal degeneration.[7,9] This led to controversy and confusion as to which of these was the primary pathologic mechanism. Some data suggested that both segmental demyelination and axonal degeneration may occur independently in alcoholic polyneuropathy.[2,24] It is now generally accepted that alcoholic polyneuropathy is an axonal degenerative neuropathy and that segmental demyelination occurs secondary to the axonal degeneration.[1,26,30]

Nerve conduction studies and needle electromyography demonstrate changes characteristic of axonal degeneration.[1,5,17,18] In most patients with alcoholic polyneuropathy, nerve conduction studies in the lower extremities disclose low amplitudes of the sensory evoked potentials,[1,5,17,18,33] mild slowing of motor conduction velocity,[1,18,33] and mild prolongation of the distal sensory and motor latencies.[1,18,33] In more severe cases, similar changes are noted in the upper extremities. Latencies of F waves and H reflexes are frequently increased or these late responses may be absent.[17,33] Abnormalities of sensory conduction, motor conduction, or late responses are commonly found in alcoholic patients who have no signs or symptoms of polyneuropathy.[3,6,13,17,31,33] Needle electromyography in alcoholic polyneuropathy usually discloses signs of axonal degeneration in a distal-proximal gradient.[1] Fibrillation potentials, positive sharp waves, and increased-duration and increased-amplitude polyphasic motor unit potentials are typical findings. Interference patterns may be decreased and are usually characterized by a reduced number of MUPs firing rapidly during attempted maximum voluntary contraction. Single fiber electromyography in alcoholic polyneuropathy discloses increased fiber density and jitter, features which are also consistent with axonal degeneration and reinnervation.[25]

The precise etiology of alcoholic polyneuropathy has been a point of continuing controversy.[27] At issue is whether alcoholic polyneuropathy is a result of nutritional deficiency or a direct toxic effect of alcohol on peripheral nerve. An overwhelming amount of evidence supports the contention that alcoholic polyneuropathy is a result of nutritional deficiency produced when chronic alcohol intake displaces essential nutrients from the diet.[19,20,22,23,27,34] It should be emphasized that the clinical, electromyographic, and pathologic features of alcoholic polyneuropathy are indistinguishable from those of neuropathic beriberi, a well established disorder of nutritional deficiency.[27] The evidence supporting a direct toxic mechanism comes from reports of polyneuropathy occurring in alcoholic patients who were apparently not malnourished[1] and from experimental animal models of alcoholism in which polyneuropathy developed in (presumably) adequately nourished animals.[4,11,12] In most of these reports, however, the criteria for adequate nutrition were not strict enough to establish that nutritional deficiencies were not factors in the pathogenesis of the polyneuropathy. It has yet to be firmly established that alcohol, acting as a neurotoxin, is the principal mechanism in the pathogenesis of alcoholic polyneuropathy.

The clinical, electromyographic, and pathologic features of alcoholic polyneuropathy are nonspecific and are very similar to other axonal degenerative polyneuropathies. The diagnosis of alcoholic polyneuropathy may be suspected when the typical features of a generalized sensorimotor axonal degenerative polyneuropathy occur in the setting of long-standing severe alcoholism and nutritional deficiency. The lack of specific diagnostic criteria for alcoholic polyneuropathy necessitates exclusion of other causes of similar clinical syndromes before a definitive diagnosis is made. Polyneuropathies resulting from disorders such as diabetes mellitus, toxic exposures, uremia, etc., may be differentiated from alcoholic polyneuropathy on the basis of typical clinical or laboratory features. Electrodiagnostic studies are not particularly helpful in differentiating these polyneuropathies.

The principal therapy for alcoholic polyneuropathy is the institution of a nutritionally balanced diet supplemented by all the B vitamins. Despite such therapy most patients show only modest improvement. The rehabilitation of the alcoholic patient is the ultimate therapeutic goal. Abstinence from alcohol is essential for the establishment of a sound nutritional state which will prevent progression or recurrence of the neuropathy.

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